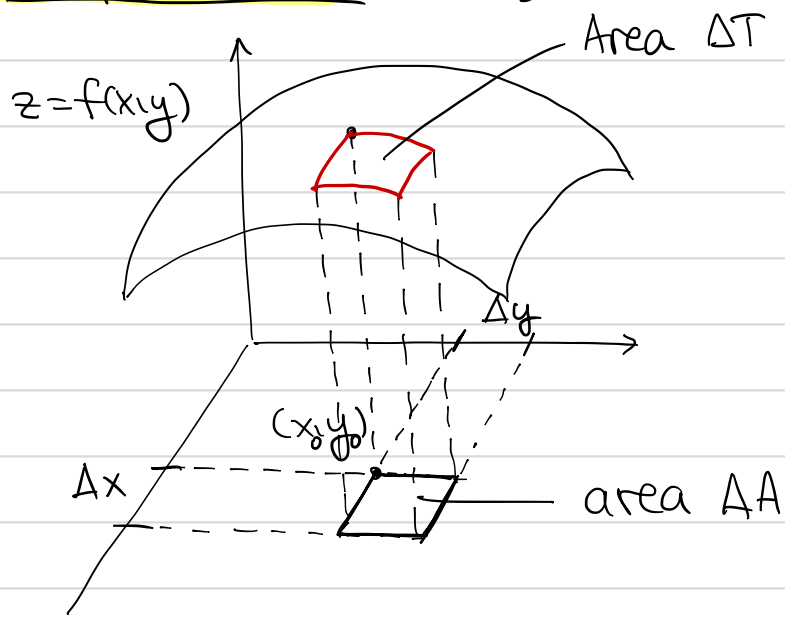
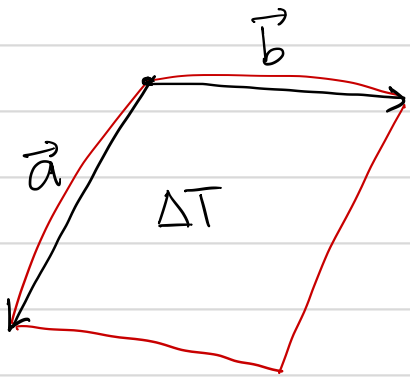
Surface area (15.5)



Zoom



$$\vec{a} = (\Delta x, 0, f(x_0+\Delta x, y_0) - f(x_0,y_0))$$
$$\approx \left(\Delta x, 0, \frac{\partial f}{\partial x}(x_0,y_0)\,\Delta x\right)$$

$$\vec{b} = (0, \Delta y, f(x_0, y_0+\Delta y) - f(x_0,y_0))$$
$$\approx \left(0, \Delta y, \frac{\partial f}{\partial y}(x_0,y_0)\,\Delta y\right)$$

$$\vec{a}\times\vec{b} \approx \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \Delta x & 0 & f_x \Delta x \\ 0 & \Delta y & f_y \Delta y \end{vmatrix}$$

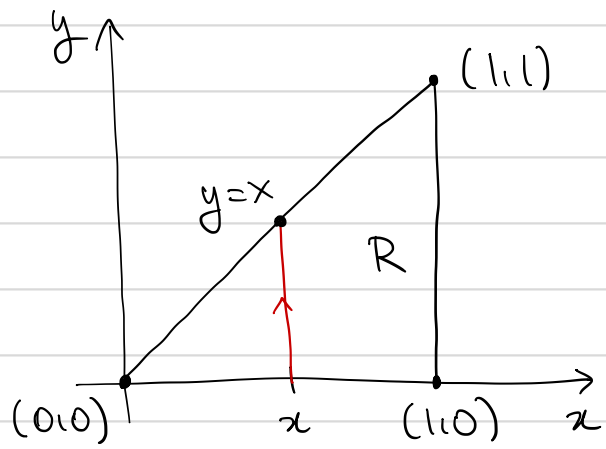
$$= (-f_x \Delta x \Delta y, -f_y \Delta x \Delta y, \Delta x \Delta y)$$

$$\Delta T \approx |\vec{a}\times\vec{b}|$$
$$\approx |(-f_x, -f_y, 1)|\,\Delta x \Delta y$$
$$= \sqrt{f_x^2 + f_y^2 + 1}\,\Delta x \Delta y$$

Area of $z=f(x,y)$ over region R of xy-plane:

$$\boxed{\text{Surface Area} = \iint_R \sqrt{f_x^2+f_y^2+1}\,dA}$$

Example: $f(x,y) = x^2 + 2y$



$$A = \iint_R \sqrt{(2x)^2 + (2)^2 + 1}\, dA$$

$$= \int_0^1 \int_0^x \sqrt{4x^2 + 5}\, dy\, dx$$

$$= \int_0^1 \sqrt{4x^2+5}\, x\, dx$$

$$= \frac{2}{3} \cdot \frac{1}{8} (4x^2+5)^{3/2} \Big|_0^1 = \frac{1}{12}(27 - 5\sqrt{5})$$